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Statistical Analysis of Risk & Returns on Varying Assets

Introduction

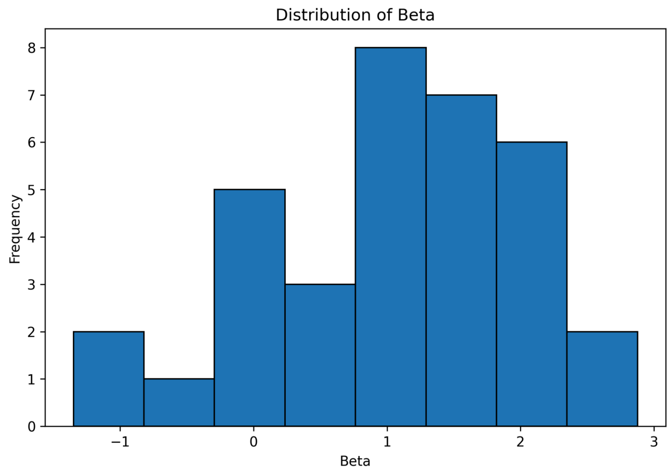
The goal of this project is to examine the relationship between risk and short-term returns across financial assets. In financial theory, investors often expect that assets with higher risk should offer higher expected returns. This project studies whether that idea appears in a small sample of U.S. stocks and major cryptocurrencies over a recent six-month period. The main question is whether higher-beta assets have higher average weekly returns than lower-beta assets.

The null hypothesis is that the mean short-term return of high-beta assets is less than or equal to the mean short-term return of low-beta assets. The alternative hypothesis is that the mean short-term return of high-beta assets is strictly greater than the mean short-term return of low-beta assets. In addition to average return, this project also considers volatility and maximum drawdown! Volatility measures how spread out weekly returns are, while maximum drawdown measures the largest percentage decline from a peak to a low point during the period. These variables are useful because an asset can have a strong average return while still exposing investors to large losses.

Data

The dataset was created using Python (and internal python libraries for data cleaning and visualization) and the yfinance library, which retrieves publicly available price data from Yahoo Finance. The sample includes 30 large U.S. stocks and four major cryptocurrencies: Bitcoin, Ethereum, XRP, and Solana. Weekly closing prices over a six-month period were collected. Then from these prices, the weekly percentage returns were calculated.

For each asset, 5 main variables were constructed. Beta measures sensitivity to market movement. For stocks, beta was calculated relative to the S&P 500. For cryptocurrencies, Bitcoin was used as the comparison asset, with Bitcoin assigned a beta a constant value of 1. Average weekly return measures the mean weekly percentage return. Volatility is the standard deviation of weekly returns. Maximum drawdown is the largest percentage loss from a prior high point during the sample. Finally, type measures whether the asset is a stock or cryptocurrency! The figure below shows the distribution of beta values across the assets in the sample. The beta values are spread across a wide range, which supports using beta as the main risk measure in this project.



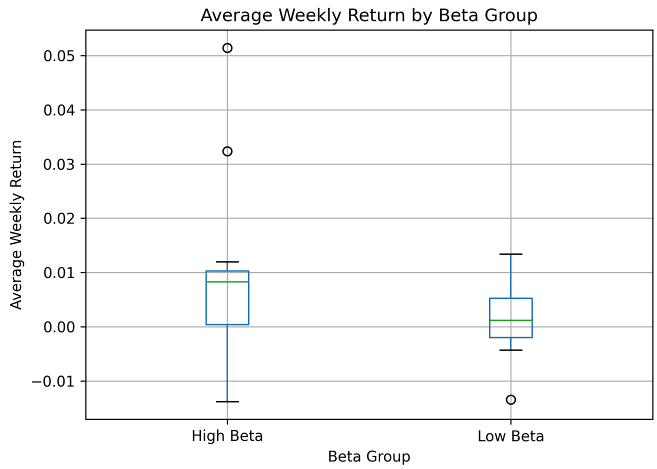
The first part of the project showed that the dataset contains a broad range of beta values and return behavior. The distribution of beta values showed that most assets were near the range of approximately 0.5 to 1.5, with some higher outliers as well. The distribution of average weekly returns was somewhat uneven, which is expected with financial data because short-term returns can be affected by market news and earnings reports.

Analysis

The first technique used was descriptive statistical analysis. Across all assets in the sample, the code computes the mean average weekly return and a 95% confidence interval. It also computes the mean weekly volatility and a 95% confidence interval. These results show whether the sample had noticeable variation in both returns and risk. The descriptive statistics also show why it is important to examine more than just average return. Some assets may appear attractive based on return, but they may also have greater volatility or larger drawdowns.

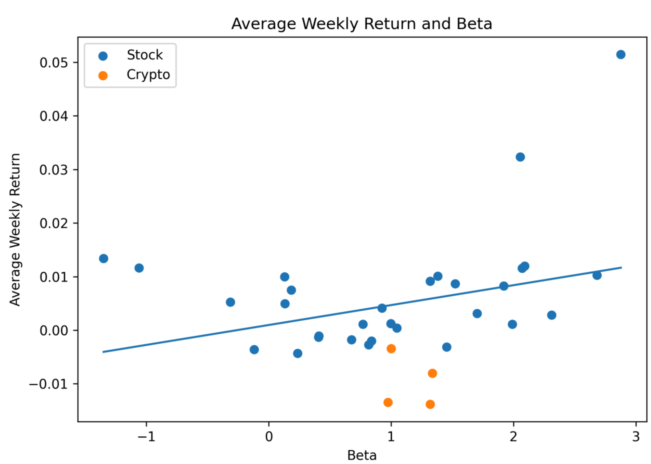
The second technique used was a Welch-two-sample t-test. Assets were split into high-beta and low-beta groups using the median beta value. This split created two groups for comparing average weekly returns. The test directly evaluates whether the high-beta group had a higher mean average weekly return than the low-beta group. The p-value from this test should be compared to a typical significance level such as 0.05. If the p-value is below 0.05, the result supports the alternative hypothesis. If the p-value is above 0.05, then the data does not provide strong enough evidence to conclude that high-beta assets had higher short-term returns. Figure 2 compares the average weekly returns of the high-beta and low-beta groups.

The high-beta group appears to have a higher median return, but it also shows more variation and several



outliers.

The third technique used was multiple linear regression. The response variable was average weekly return, and the explanatory variables were beta, volatility, maximum drawdown, and an indicator variable for whether the asset was a cryptocurrency. The purpose of this model was to test whether beta still helps explain average return after accounting for other risk-related variables. The R-squared value measures how much of the variation in average return is explained by the model. The beta coefficient shows whether higher beta is associated with higher or lower average return after controlling for the other variables. Figure 3 provides a visual summary of the relationship between beta and average weekly return. The fitted line slopes upward, suggesting a positive relationship, but the scatter of the points shows that beta alone does not fully explain short-term return.



The regression results should be interpreted carefully because the dataset is relatively small and covers only a six-month period. Short-term financial returns are noisy, and a few unusually strong or weak assets can affect the results. The residual plot was included to check whether the model had obvious issues such as extreme residual patterns. If the residuals appear randomly scattered around zero, then the linear model is more reasonable. If the residuals show a curved pattern or several extreme outliers, then the model may not fully capture the behavior of the data.

Conclusion

In conclusion, this project tested whether higher-risk assets had higher short-term returns in a sample of stocks and cryptocurrencies. The descriptive statistics showed that the assets varied in beta, volatility, average return, and maximum drawdown. The t-test compared high-beta and low-beta assets directly, while the regression model looked at whether beta helped explain average return after accounting for volatility, drawdown, and asset type.

The main conclusion is that higher beta may be connected to higher returns, but the relationship is not completely clear over a short and strict six-month period (lacking diversity!). Short-term returns can be affected by market news, investor behavior, earnings results, and crypto-specific volatility. The project also has some limitations, including the small sample size, short time period, and the use of Bitcoin as a simple proxy for the crypto market. Therefore, even with these limits, the analysis shows that beta alone is not enough to evaluate an asset, and investors should also consider volatility and downside risk.

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